Fish and shellfish allergy in children: review of a persistent food allergy.

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The increased consumption of fish and shellfish has resulted in more frequent reports of adverse reactions to seafood, emphasizing the need for more specific diagnosis and treatment of this condition and exploring reasons for the persistence of this allergy. This review discusses interesting and new findings in the area of fish and shellfish allergy. New allergens and important potential cross-reacting allergens have been identified within the fish family and between shellfish, arachnids, and insects. The diagnostic approach may require prick to-prick tests using crude extracts of both raw and cooked forms of seafood for screening seafood sensitization before a food challenge or where food challenge is not feasible. Allergen-specific immunotherapy can be important; mutated less allergenic seafood proteins have been developed for this purpose. The persistence of allergy because of seafood proteins' resistance after rigorous treatment like cooking and extreme pH is well documented. Additionally, IgE antibodies from individuals with persistent allergy may be directed against different epitopes than those in patients with transient allergy. For a topic as important as this one, new areas of technological developments will likely have a significant impact, to provide more accurate methods of diagnosing useful information to patients about the likely course of their seafood allergy over the course of their childhood and beyond.